# COMPARATIVE ANALYSIS OF CANCER INCIDENCE PATTERNS ACROSS POPULATION-BASED CANCER REGISTRIES IN INDIA (1985–2019)

**A Project Submitted to Bharathiar University in Partial Fulfilment of the Requirements for the Award of the Degree of**

**MASTER OF SCIENCE IN BIOSTATISTICS**

Submitted by


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# DECLARATION

I **Gokul B (24STAD006)** hereby declare that the Project work entitled **" Comparative Analysis of Cancer Incidence Patterns across Population-Based Cancer Registries in India (1985–2019) "** submitted to the Bharathiar University, Coimbatore, in partial fulfilment of the requirements for the award of the Degree of Master of Science in Biostatistics is a record of original research work done by me under the supervision and guidance of **Dr. S. JAYALAKSHMI**, Assistant Professor, Department of Statistics, Bharathiar University, Coimbatore. This Project work has not formed the basis for the award of any Degree, Diploma, Associateship, Fellowship or other similar title to any candidate of any University.

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**(Dr. S. JAYALAKSHMI)**

# CERTIFICATE

This is to certify that the Project work entitled **" Comparative Analysis of Cancer Incidence Patterns across Population-Based Cancer Registries in India (1985–2019) "**submitted to Bharathiar University, Coimbatore, in partial fulfilment of the requirements for the award of the Degree of Master of Science in Biostatistics, is a record of original research work done by **Gokul B (24STAD006)** during the period of his study in the Department of Statistics, Bharathiar University, under my supervision and guidance and the Project work has not formed the basis for the award of any Degree / Diploma/ Associateship/Fellowship or other similar title to any candidate of any University.

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# BIOSTATISTICS

## 1.1 Introduction

Biostatistics refers to the application of statistical principles and methods to biological, medical, and public health data. It enables researchers to design studies, analyze complex datasets, interpret findings, and draw meaningful conclusions from empirical observations. Epidemiology, on the other hand, is the study of the distribution and determinants of health-related states or events in specified populations and the application of this knowledge to control health problems.

Public health is a multidisciplinary field concerned with the protection and improvement of the health of populations through preventive measures, health promotion, and disease control. Among the core disciplines that support public health research and practice, biostatistics and epidemiology play a fundamental role. These fields provide the methodological and analytical frameworks necessary for understanding patterns of disease occurrence, identifying determinants of health outcomes, and evaluating interventions designed to improve population health. Together, they form the scientific backbone of evidence-based public health.

**Figure 1.1 Biostatistics and Epidemiology**

In modern public health research, the integration of epidemiological concepts with advanced statistical methods has become increasingly important. With the growing availability of large-scale health datasets, electronic health records, and population-based surveillance systems, biostatistical techniques such as survival analysis, regression modeling, and machine learning are widely used to explore disease risk factors and predict health outcomes. Epidemiological approaches, including cohort studies, case-control studies, and cross-sectional

surveys, provide structured methodologies for collecting and interpreting data related to disease occurrence.

The importance of biostatistics and epidemiology has been particularly evident in the study of chronic diseases such as cancer, cardiovascular diseases, and diabetes, which represent a major burden on global health systems. These disciplines allow researchers to measure disease incidence, prevalence, mortality, and survival patterns across populations. Moreover, they support the evaluation of prevention strategies, screening programs, and treatment interventions.

Therefore, the integration of biostatistics and epidemiology is essential for generating reliable evidence that informs public health policies and clinical decision-making.

## 1.2 Biostatistics in Public Health

### 1.2.1 Definition and Scope of Biostatistics

Biostatistics is a branch of statistics that deals with the application of statistical methods to biological, medical, and health-related data. It plays a critical role in the design, analysis, and interpretation of health research. Biostatistical techniques help transform raw data into meaningful information that can guide medical practice, health policy decisions, and disease prevention strategies.

### 1.2.2 Role of Biostatistics in Study Design

One of the most important contributions of biostatistics to public health research is study design. Proper study design ensures that research findings are valid, reliable, and generalizable. Biostatisticians work closely with epidemiologists and researchers to determine sample size, sampling methods, study duration, and outcome variables.

A well-designed study reduces bias and improves the precision of statistical estimates. It also ensures that data collected during the research process are suitable for statistical analysis.

### 1.2.3 Descriptive Statistics in Public Health

Descriptive statistics are used to summarize and present health-related data. Measures such as mean, median, mode, standard deviation, and percentages provide insights into the distribution and characteristics of populations.

For example, descriptive statistics may be used to summarize patient age distribution, gender distribution, incidence rates of diseases, and other demographic characteristics in epidemiological studies.

### 1.2.4 Inferential Statistics in Health Research

Inferential statistical methods allow researchers to draw conclusions about populations based on sample data. Techniques such as hypothesis testing, confidence intervals, and regression analysis are widely used in public health studies.

Regression models, including logistic regression and linear regression, help identify relationships between risk factors and disease outcomes. These methods enable researchers to determine whether observed associations are statistically significant.

### 1.2.5 Survival Analysis in Medical Research

Survival analysis is an important statistical method used in clinical and epidemiological research to analyze **time-to-event data**. It is commonly applied in studies of chronic diseases such as cancer to estimate survival probabilities and identify factors influencing mortality.

Methods such as the Kaplan–Meier estimator and Cox proportional hazards model are frequently used to analyze survival data and evaluate treatment outcomes.

### 1.2.6 Modern Developments in Biostatistics

With advances in computational technology, biostatistics has expanded to include modern analytical techniques such as Bayesian statistics, machine learning, and high-dimensional data analysis. These methods are increasingly applied in genomic research, personalized medicine, and predictive modeling of disease risk.

## 1.3 Epidemiology in Public Health

### 1.3.1 Definition and Concept of Epidemiology

Epidemiology is the scientific study of the distribution and determinants of health-related states or events in populations. It focuses on identifying the causes of diseases and applying this knowledge to control health problems.

Epidemiology serves as the foundation of public health because it provides the evidence needed to develop effective prevention and control strategies.

### 1.3.2 Measures of Disease Frequency

Understanding disease frequency is a central component of epidemiological research. Two important measures are **incidence** and **prevalence**.

Incidence refers to the number of new cases of a disease occurring within a specific time period, while prevalence represents the total number of existing cases in a population at a given time.

These measures help researchers assess the burden of diseases and identify trends in population health.

### 1.3.3 Determinants of Disease

Epidemiologists investigate various factors that influence the occurrence of diseases. These determinants include biological factors, environmental exposures, lifestyle behaviors, and socioeconomic conditions.

For example, tobacco consumption has been identified as a major risk factor for lung cancer and other respiratory diseases through epidemiological research.

### 1.3.4 Epidemiological Study Designs

Several study designs are used in epidemiology to investigate disease patterns and risk factors.

### 1.3.5 Cross-Sectional Studies

Cross-sectional studies examine the relationship between exposure and disease at a specific point in time. A cross-sectional study is an observational study design in which data on exposure and outcome are collected simultaneously at a single point in time or over a short period. It provides a “snapshot” of the health status of a population and is widely used in epidemiology to assess the prevalence of diseases and associated risk factors.

Unlike longitudinal studies, cross-sectional studies do not involve follow-up over time. Therefore, they are primarily useful for describing the burden of disease rather than establishing causal relationships.

### 1.3.6 Case-Control Studies

Case-control studies compare individuals with a disease to those without the disease (controls) to identify potential risk factors. A case-control study is an observational analytical study design in which individuals with a specific disease or outcome (cases) are compared to

individuals without the disease (controls) to identify factors that may have contributed to the occurrence of the disease. This design is particularly useful for studying rare diseases and conditions with long latency periods.

Unlike cohort studies, case-control studies begin with the outcome and look backward to assess exposure history. This retrospective nature allows for efficient investigation of potential risk factors.

### 1.3.7 Cohort Studies

Cohort studies follow groups of individuals over time to examine the relationship between exposures and disease outcomes. A cohort study is an observational analytical study design in which a group of individuals (cohort) is followed over time to assess the relationship between exposure to a risk factor and the occurrence of a specific outcome or disease. Participants are classified based on their exposure status, and the incidence of disease is compared between exposed and unexposed groups.

Cohort studies are fundamentally longitudinal, allowing for the establishment of a temporal relationship between exposure and outcome, which is critical for causal inference.

### 1.3.8 Disease Surveillance

Disease surveillance involves the continuous monitoring and analysis of disease occurrence within populations. Surveillance systems help identify outbreaks, monitor disease trends, and evaluate public health interventions.

Population-based cancer registries are an example of surveillance systems that collect data on cancer incidence, mortality, and survival.

### 1.3.9 Epidemiology of Non-Communicable Diseases

In recent decades, epidemiological research has increasingly focused on non-communicable diseases (NCDs) such as cancer, cardiovascular diseases, and diabetes. These diseases represent a major public health challenge worldwide and require long-term monitoring and prevention strategies.

### 1.3.10 Integration of Biostatistics and Epidemiology

Biostatistics and epidemiology are closely interconnected disciplines that complement each other in public health research. Epidemiology provides the conceptual framework for

studying disease patterns, while biostatistics offers the analytical tools needed to evaluate these patterns quantitatively.

The integration of these disciplines allows researchers to design scientifically sound studies and conduct rigorous data analysis. Advanced statistical methods such as multivariable regression models, survival analysis, and time-series analysis are widely used in epidemiological research to identify risk factors and assess disease outcomes.

### 1.3.11 Role in Disease Surveillance and Prevention

Biostatistics and epidemiology play an essential role in monitoring disease trends and implementing effective prevention strategies. Statistical analysis of surveillance data helps identify high-risk populations and evaluate the effectiveness of public health programs.

In cancer research, population-based cancer registries provide valuable data on cancer incidence, mortality, and survival patterns. These data help guide screening programs, treatment strategies, and prevention initiatives.

Preventive measures such as vaccination programs, tobacco control policies, and health education campaigns are often developed based on epidemiological evidence supported by statistical analysis.

### 1.3.12 Importance in Evidence-Based Public Health

Evidence-based public health involves the use of scientific evidence to guide policy decisions and healthcare practices. Biostatistics provides the analytical tools required to evaluate health data, while epidemiology offers the theoretical framework for interpreting research findings.

The integration of these disciplines ensures that public health interventions are based on reliable evidence. Statistical models and epidemiological analyses help predict disease trends, assess intervention outcomes, and guide health policy decisions aimed at improving population health.

# LEADING CANCER SITES

Cancer is a disease in which there is unregulated cell growth in any organ systems occurring in humans of all age groups, irrespective of age, sex, nationality, ethnicity, economic status, educational strata, geological and geographic distributions. In its presentation, it could be acute (sudden onset), sub-acute (slow onset), or chronic (long period of time). In its symptoms, it is known to present itself in the most insidious non-specific presenting symptoms like fever, diarrhoea or weight loss to the symptoms like bleeding, obstructive symptoms, growths. As a disease it has the potential to restrain a person from achieving full physical, physiological, psychological and economic potential. It's a major concern for the patient, his/her family, the clinician, the healthcare provider and the taxpayer. The aetiology of cancer is multi-factorial: genetic predisposition, exposure to tobacco, certain chemicals, infections, radiation, inappropriate lifestyle factors (alcohol, Inappropriate diet, physical inactivity, high body mass index, diabetics and metabolic syndrome) have all been implicated in the causation of cancer. Yet, the cause remains undetermined in a large proportion of patients. This is important since much of the preventive strategies are based on some of the known factors. Associations have also been made on the degree of exposure, dose of exposure, duration of exposure, age (vulnerability) of the exposed and the like.

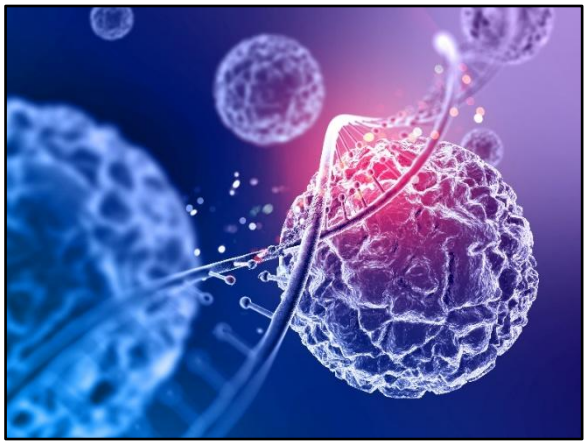


***Figure 2.1 Cancer Cells***

All these factors have come from deductive reasoning of epidemiological data and an insight into the possible causative mechanisms. There have been remarkable successes in the evolving treatment modalities which have strengthened the fight against cancer.

## 2.1 Global Burden of Cancer

Cancer is one of the leading causes of morbidity and mortality worldwide and represents a major public health challenge in both developed and developing countries. According to the World Health Organization, cancer accounted for approximately 10 million deaths globally in recent years, making it one of the most significant contributors to premature mortality and disability. The global cancer burden continues to increase due to population growth, ageing populations, urbanization, and the increasing prevalence of behavioural and environmental risk factors such as tobacco use, unhealthy diet, physical inactivity, alcohol consumption, and exposure to environmental carcinogens.

***Figure 2.2 IARC - WHO***

The International Agency for Research on Cancer estimates that millions of new cancer cases are diagnosed annually worldwide. As life expectancy increases and populations age, the incidence of cancer is expected to rise substantially in the coming decades. The global distribution of cancer is not uniform; rather, it varies considerably across countries and regions due to differences in socioeconomic development, lifestyle patterns, environmental exposures, and access to healthcare services.

In many high-income countries, cancers such as lung, colorectal, breast, and prostate cancers represent a major share of the disease burden. In contrast, several low- and middle-income countries continue to experience high incidence of infection-related cancers such as cervical, liver, and stomach cancers. This variation reflects the complex interaction of biological, behavioural, and environmental determinants that influence cancer risk.

## 2.2 Cancer Burden in India

India is currently experiencing a significant epidemiological transition characterized by a growing burden of non-communicable diseases, including cancer. Rapid urbanization, demographic ageing, lifestyle changes, and environmental exposures have contributed to a

steady rise in cancer incidence across the country. Data compiled by the Indian Council of Medical Research indicate that cancer incidence and mortality are increasing in many regions of India.

Cancer patterns in India demonstrate considerable geographic and demographic variation. Certain cancers such as oral, lung, and esophageal cancers are particularly common among males, largely due to the widespread use of tobacco in both smoked and smokeless forms. Among females, breast and cervical cancers represent the most frequently diagnosed malignancies. The rising incidence of breast cancer in urban populations and the persistence of cervical cancer in many regions highlight the dual burden of lifestyle-related and infection-related cancers in the country.

To systematically monitor cancer trends, India established the National Cancer Registry Programme, which collects data through Population-Based Cancer Registries (PBCRs) and Hospital-Based Cancer Registries (HBCRs). These registries provide valuable information on cancer incidence, mortality, and survival patterns across different geographic regions. PBCR data are particularly important for understanding population-level cancer patterns and guiding public health policy.

### 2.3 The National Cancer Registry Programme

Cancer registry is an organization of systematic collection, storage, analysis, interpretation and reporting of data on patients with cancer (IARC). A proper analysis and interpretation of data provide insights with inputs for its prevention, control and management. Time-trend studies are also possible when data have been accumulated over long periods of time. In addition to incidence figures, population-based cancer registries who conduct follow-up of their patients are able to estimate the prevalence of cancer. A cancer registry provides an economical and efficient method of ascertaining cancer occurrence rather than intervention trials and cohort studies. In India, the National Cancer Registry Programme (NCRP) under the Indian Council of Medical Research (ICMR) with its network of cancer registries was started in December 1981 with the co-ordinating centre at Bengaluru. Presently it is operated by the ICMR-NCDIR, Bengaluru. This provides the data on cancer incidence, mortality, pattern, trend and geo-pathological distribution of cancers. It also helps to formulate and implement policies and programmes, monitor and evaluate the cancer control activities.

***Figure 2.3 NCRP***

There are two types of cancer registries under the programme. Population Based Cancer Registries (PBCRs) record all the new cancer cases occurring in a denied population within a geographic area. The Hospital Based Cancer Registries (HBCRs) record information on cancer patients attending a particular hospital, with focus on clinical care, treatment and outcome. Cancer Atlas approaches have also been used for specific short-term purposes.

The main objectives of the programme are:

1. To generate reliable data on the magnitude and patterns of cancer.
2. Propose further epidemiological studies based on results of registry data.
3. Help in designing, planning, monitoring and evaluation of cancer control activities under the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS).
4. Develop training programmes in cancer registration and epidemiology.

NCRP started with a network of three PBCRs in Bangalore, Chennai and Mumbai and three HBCRs in Chandigarh, Dibrugarh and Thiruvananthapuram. The number of registries working under the programme have expanded greatly from the time of inception and presently there are 36 PBCRs and 236 HBCRs registered under NCRP.

Since cancer is not a notifiable disease, cancer registration in India is active and staff of all registries visit hospitals, pathology laboratories and all other sources of registration of cancer cases on a routine basis. Death certificates are also scrutinized from the local government units like municipal corporation and panchayat raj institutes and information is collected on all cases where cancer is mentioned as a cause of death on the death certificates.

## 2.4 Population-Based Cancer Registries (PBCRs)

Population-Based Cancer Registries play a crucial role in cancer surveillance by systematically collecting and analyzing data on all newly diagnosed cancer cases occurring within a defined population. The information generated by PBCRs enables researchers and policymakers to assess the magnitude and distribution of cancer, evaluate temporal trends, and identify populations at increased risk.

PBCRs in India operate under the framework of the National Cancer Registry Programme and cover several urban and rural populations across the country. Registries such as Chennai, Bangalore, Mumbai, Delhi, Bhopal, and Barshi Rural have provided long-term datasets that are valuable for studying cancer epidemiology in India. These registries record detailed information on patient demographics, cancer site, stage at diagnosis, and survival outcomes.

The availability of PBCR data has enabled researchers to analyze cancer incidence patterns, evaluate regional variations, and assess changes in cancer burden over time. Such analyses are essential for planning cancer prevention and control strategies, allocating healthcare resources, and monitoring the effectiveness of public health interventions.

## 2.5 Role of Biostatistics in Cancer Epidemiology

Biostatistics plays a fundamental role in cancer research by providing quantitative methods for analyzing and interpreting complex biomedical data. Statistical techniques allow researchers to evaluate associations between risk factors and disease occurrence, assess variations in incidence rates across populations, and estimate survival outcomes among cancer patients.

Descriptive statistical methods are commonly used to summarize cancer incidence, prevalence, and mortality patterns across populations. Measures such as relative proportions, crude rates, and age-adjusted incidence rates provide insight into the distribution of cancer across different demographic groups. Age standardization is particularly important in cancer epidemiology because cancer risk increases substantially with age, and populations may differ significantly in their age structures.

In addition to descriptive analysis, survival analysis methods are widely used to evaluate cancer prognosis. Techniques such as the Kaplan–Meier method enable estimation of survival probabilities over time, while regression models such as the Cox proportional hazards

model allow researchers to identify factors that influence survival outcomes. These statistical tools provide critical insights into disease progression and treatment effectiveness.

The integration of biostatistical methods with cancer registry data enables researchers to explore patterns of disease occurrence, identify risk factors, and evaluate interventions aimed at reducing cancer burden.

### 2.6 Median Age at Cancer Diagnosis

Age is one of the most important determinants of cancer occurrence. The probability of developing cancer increases substantially with advancing age due to cumulative exposure to risk factors, biological ageing processes, and genetic mutations that accumulate over time. As a result, the age distribution of cancer cases provides valuable information about disease patterns within populations.

Median age at diagnosis is a useful summary measure that represents the central tendency of age distribution among cancer patients. Unlike mean age, the median is less affected by extreme values and provides a more robust indicator of the typical age at which cancer is diagnosed within a population.

In epidemiological studies, both crude median age and age-adjusted median age are often used. Crude median age reflects the observed age distribution within the population, whereas age-adjusted median age accounts for differences in population age structures. The comparison of these measures helps researchers understand whether observed differences in cancer patterns are due to demographic variations or underlying epidemiological factors.

Analyzing trends in median age at diagnosis over time can also reveal important information about shifts in cancer epidemiology. For example, increasing median age at diagnosis may reflect improvements in life expectancy, delayed onset of disease, or changes in screening and diagnostic practices.

### 2.7 Importance of Survival Analysis in Cancer Studies

Survival analysis is a key component of cancer epidemiology because it allows researchers to evaluate patient outcomes following diagnosis. Unlike many other diseases, cancer often involves long periods between diagnosis and death, making time-to-event analysis essential for understanding prognosis and treatment effectiveness.

The Kaplan–Meier method is widely used to estimate survival probabilities over time, taking into account censored observations where the outcome event has not occurred during

the follow-up period. This method provides a graphical representation of survival patterns and allows comparisons between different patient groups.

The Cox proportional hazards model is another important statistical tool used to evaluate the effect of multiple predictors on survival outcomes. By estimating hazard ratios, the Cox model quantifies the relative risk of death associated with specific variables such as age, sex, stage of disease, or treatment factors. These analyses help identify prognostic factors that influence survival and guide clinical decision-making.

Understanding survival patterns is also important for evaluating the effectiveness of cancer control programs and healthcare systems. Improved survival outcomes may indicate advances in early detection, treatment technologies, and patient management strategies.

### 2.8 Rationale for the Study

Despite considerable progress in cancer surveillance in India, significant regional variations in cancer incidence and survival patterns remain poorly understood. Differences in lifestyle factors, environmental exposures, healthcare accessibility, and diagnostic practices may contribute to variation in cancer burden across regions.

Analyzing data from multiple PBCRs provides an opportunity to examine these variations systematically. By comparing cancer incidence patterns, median age at diagnosis, and survival outcomes across different registries, it is possible to gain a deeper understanding of the epidemiological dynamics of cancer in India.

Furthermore, the application of advanced statistical methods such as survival analysis enables the identification of factors that influence patient prognosis. Such insights are essential for improving cancer prevention strategies, optimizing screening programs, and enhancing treatment outcomes.

### 2.9 Aim of the Study

The present study aims to analyze cancer incidence patterns, median age at diagnosis, and survival outcomes across selected Population-Based Cancer Registries in India using statistical methods.

### 2.10 Significance of the Study

The findings of this study are expected to contribute to a better understanding of cancer epidemiology in India. By examining trends in median age at diagnosis and variations in cancer

incidence across registries, the study provides valuable information for researchers, clinicians, and public health policymakers.

In addition, the survival analysis component of the study offers insights into prognostic factors that influence patient outcomes. These findings can support the development of targeted interventions aimed at improving early detection, treatment, and overall cancer control strategies.

Ultimately, strengthening the integration of biostatistics with cancer epidemiology will enhance the capacity of healthcare systems to respond effectively to the growing burden of cancer.

### 2.11 Objectives of the Study

- To identify the seven leading cancer sites among males and females and estimate their relative proportions among the total registered cancer cases.
- To analyze the burden and regional distribution of cancer incidence among males and females across selected Population-Based Cancer Registries (PBCRs).
- To examine temporal trends in crude and age-adjusted median age at cancer diagnosis across six Population-Based Cancer Registries (PBCRs) in India during the period 1985–2019.
- To evaluate the influence of sex as a prognostic factor on cancer survival using survival analysis methods.
- To assess the effect of increasing age on mortality risk among cancer patients using appropriate statistical models.

# REVIEW OF LITERATURE

**Zahed et al**. (2024) conducted **"An International comparative study on age at diagnosis for lung, colon, breast, and prostate cancers"** using data from population-based cancer registries in 63 countries obtained from the Cancer Incidence in Five Continents (CI5) database maintained by the International Agency for Research on Cancer.

**Findings:** The study used median age estimation, indirect age standardization, Poisson regression modelling, and correlation analysis to examine variations in cancer diagnosis age across countries. The results showed that crude differences in median age at diagnosis across countries were as high as 20 years, but after adjusting for population age structure the variation reduced to approximately 10–12 years. Colon cancer was diagnosed at younger ages in low- and middle-income countries, whereas prostate cancer tended to occur at older ages. The study concluded that differences in cancer diagnosis age are influenced by population structure, risk factor prevalence, and screening practices.

**Sathishkumar et al.** (2020) evaluated "**The impact of using the Indian Standard Population (ISP) for estimating Age Adjusted Rates in Population-Based Cancer Registries in India".** Using direct age standardization.

**Findings:** The study compared incidence rates calculated using the Segi World Standard Population and the Indian Standard Population. The results indicated that the use of ISP resulted in approximately 20% lower Age Adjusted Rates, while the relative ranking of cancer registries remained mostly unchanged. The authors suggested that ISP provides a more realistic representation of the Indian population structure for comparing cancer incidence within the country.

**Alshamsan et al.** (2026) conducted **"A population-based study to analyze trends in breast cancer incidence, age at diagnosis, and stage distribution in Saudi Arabia using data from the Saudi Cancer Registry from 2002 to 2022".**

**Findings:** The study applied age-standardized incidence rates (ASR), descriptive epidemiological analysis, and Join point regression modeling to estimate annual percent change (APC) and average annual percent change (AAPC). The results showed that the ASR increased from 12.6 to 49.7 per 100,000 population, indicating a significant rise in breast cancer incidence (AAPC 5.6%, $p < 0.0001$). The median age at diagnosis increased from 46 years to

52 years, and the stage distribution shifted toward earlier detection, with localized disease increasing from 26% to 52% over the study period. The study concluded that breast cancer incidence has increased rapidly in Saudi Arabia and emphasized the importance of expanding screening and early detection programs.

**Sathishkumar et al.** (2022) conducted a national study titled "**Cancer Incidence Estimates for 2022 and Projection for 2025 in India**" using data from 28 Population-Based Cancer Registries under the National Cancer Registry Programme (NCRP).

**Findings:** The study used age-specific incidence rates (ASIR), crude rates (CR), and age-adjusted rates (AAR) derived from PBCR data for 2012–2016 and applied them to population estimates from the 2001 and 2011 Indian census to estimate cancer incidence in 2022. The study estimated 1,461,427 new cancer cases in India in 2022, with a crude incidence rate of 100.4 per 100,000 population. Breast cancer was the most common cancer among females (28.8%), while lung cancer was the leading cancer among males (10.6%). The study also reported that one in nine individuals in India is likely to develop cancer during their lifetime, and cancer incidence is projected to increase by 12.8% by 2025.

**Sathishkumar et al**. (2024) conducted **"A population-based study on breast cancer survival using 17,331 cases diagnosed between 2012 and 2015 from 11 Population-Based Cancer Registries in India".**

**Findings:** Survival time was calculated from the date of diagnosis to death or censoring. The study estimated observed survival using the actuarial method and relative survival using the Ederer-II approach. The results showed that the overall 5-year age-standardized relative survival was 66.4%. Higher survival rates were reported in Mizoram (74.9%), Ahmedabad urban (72.7%), and Kollam (71.5%), while lower survival was observed in Pasighat (41.9%) and Tripura (45.6%). Survival was substantially higher for localized disease (81.0%) compared with distant metastasis (18.3%), indicating that early detection and timely treatment play a critical role in improving breast cancer outcomes in India.

# RESEARCH METHODOLOGY

## 3.1 Study Design

The present study adopts a retrospective observational study design based on secondary data obtained from population-based cancer registries. The objective of the study is to examine cancer incidence patterns, median age at diagnosis, and survival outcomes across selected Population-Based Cancer Registries (PBCRs) in India. Retrospective epidemiological analysis is widely used in cancer research to evaluate disease distribution and temporal trends using existing registry data.

The study integrates descriptive statistical methods and survival analysis techniques to explore variations in cancer burden across different regions and demographic groups. Descriptive methods are used to summarize cancer incidence patterns, while inferential statistical models are employed to assess survival outcomes and identify prognostic factors.

## 3.2 Data Source

The data used in this study were obtained from the National Cancer Registry Programme under the Indian Council of Medical Research. The National Cancer Registry Programme collects cancer incidence data through a network of Population-Based Cancer Registries that systematically record all new cancer cases occurring within defined populations. PBCRs provide epidemiological indicators including age-specific incidence rates, age-adjusted incidence rates, relative proportions of cancer sites, and survival outcomes. These data enable researchers to evaluate cancer trends, regional variations, and determinants of cancer burden.

## 3.3 Study Area and Population

The study includes data from selected PBCRs representing both urban and rural populations across India. The registries included in the analysis are Chennai, Bangalore, Mumbai, Delhi, Bhopal, and Barshi Rural.

These registries provide long-term cancer incidence data and represent diverse demographic and geographic populations. Both male and female cancer cases registered during the study period were included in the analysis to examine sex-specific cancer patterns.

### 3.4 Study Period

The analysis covers cancer registry data from **1985 to 2019**, depending on the availability of data across registries. For comparative analyses of incidence indicators, the consolidated PBCR reporting period **2015–2019** was considered. This period provides the most recent standardized estimates of cancer burden across registries.

### 3.5 Variables Included in the Study

The variables considered in the study include demographic, clinical, and epidemiological indicators.

**Demographic variables**

- Age at diagnosis
- Sex

**Clinical variables**

- Cancer site
- ICD–10 classification
- Date of diagnosis
- Follow-up status

**Epidemiological indicators**

- Relative proportion of cancer sites
- Age-adjusted incidence rate
- Age-specific incidence rate
- Median age at diagnosis
- Survival time

These variables enable comprehensive statistical analysis of cancer incidence patterns and survival outcomes.

### 3.6 Graphical Representation

#### 3.6.1 Bar Chart

A bar chart is a graphical representation used to display and compare categorical data using rectangular bars, where the height or length of each bar corresponds to the magnitude of the variable. It is particularly useful for summarizing discrete data such as frequencies, percentages, or relative proportions. In epidemiological studies, bar charts are commonly used to present the distribution of diseases across categories such as cancer sites, sex, or geographic

regions. The clear visual separation between bars allows easy comparison across groups. Bar charts can be presented in vertical or horizontal formats depending on the nature of the data. Overall, they provide a simple and effective way to communicate categorical differences.

### 3.6.2 Dot Plot

A dot plot is a graphical method that represents individual data points as dots along a numerical scale or axis. It is useful for displaying the distribution and variation of data across different groups or categories. Unlike bar charts, dot plots do not aggregate data, allowing each observation to be visualized independently. In biostatistics, dot plots are often used to compare values such as incidence rates or median ages across different populations or registries. This method helps in identifying patterns, clusters, and outliers more clearly. Dot plots are particularly effective when dealing with small to moderate-sized datasets and provide a clean and uncluttered visualization.

### 3.6.3 Line Plot

A line plot is a graphical representation in which data points are connected by straight lines to illustrate trends over time or across ordered categories. It is widely used in statistical and epidemiological analyses to study temporal changes in variables such as incidence rates, mortality, or median age at diagnosis. The continuity of the line helps in identifying increasing or decreasing trends and fluctuations over a period. Line plots are especially useful for time-series data and longitudinal studies. They allow easy comparison of multiple groups when more than one line is plotted. Overall, line plots provide a clear visualization of patterns and trends in data.

## 3.7 Statistical Methods

A combination of descriptive epidemiological measures and survival analysis techniques was used to examine cancer incidence patterns and survival outcomes across the selected Population-Based Cancer Registries. Statistical measures such as relative proportions, median age at diagnosis, age-specific incidence rates, age-adjusted incidence rates, and survival analysis methods were employed to provide a comprehensive understanding of cancer distribution and survival patterns. These methods allow meaningful comparison of cancer occurrence across different populations and time periods while accounting for variations in demographic characteristics.

### 3.7.1 Relative Proportion Analysis

Relative proportion analysis was used to determine the leading cancer sites among males and females across the selected registries.

The relative proportion of a cancer site was calculated using the formula:

$$\textbf{Relative Proportion (\%)} = \frac{\textbf{Number of cases of a specific cancer site}}{\textbf{Total number of cancer cases}} \times 100$$

This measure provides information on the distribution of different cancer types within the population.

### 3.7.2 Median Age at Diagnosis

Median age represents the central value in the ordered distribution of age at diagnosis. Median age at diagnosis was used as a summary measure to represent the central tendency of age distribution among cancer patients. Both **crude median age** and **age-adjusted median age** were examined.

Crude median age reflects the observed age distribution of cancer cases within the population, while age-adjusted median age accounts for variations in population age structure. The comparison of crude and adjusted median ages allows more meaningful comparison across populations with different demographic compositions.

Trend analysis of median age was conducted across successive time periods to examine changes in age distribution of cancer cases over time.

If $\boldsymbol{n}$ observations are arranged in ascending order:

$$\textbf{Median} = \begin{cases} x_{\frac{n+1}{2}}, & \textbf{if } \boldsymbol{n} \textbf{ is odd} \\ \dfrac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}, & \textbf{if } \boldsymbol{n} \textbf{ is even} \end{cases}$$

Median age was used to summarize the typical age at which cancer is diagnosed among patients.

### 3.7.3 Age-Specific Incidence Rate (ASIR)

Age-Specific Incidence Rate measures the occurrence of cancer in a particular age group relative to the population at risk in that age group.

$$\boldsymbol{ASIR} = \frac{\textbf{Number of new cancer cases in a specific age group}}{\textbf{Population in that age group}} \times 100{,}000$$

This rate helps identify age groups with higher cancer risk.

### 3.7.4 Age-Adjusted Incidence Rate (AAR)

The Age-Adjusted Incidence Rate was used to compare cancer incidence across registries while controlling for differences in population age structures. Age adjustment was performed using standard population methods commonly applied in cancer epidemiology.

AAR provides a standardized measure that allows meaningful comparison of cancer incidence between different geographic regions.

The Age-Adjusted Incidence Rate is calculated using the direct standardization method:

$$\boldsymbol{AAR} = \frac{\sum(\boldsymbol{r_i} \times \boldsymbol{w_i})}{\sum \boldsymbol{w_i}}$$

Where:

$\boldsymbol{r_i}$= age-specific incidence rate in age group $i$

$\boldsymbol{w_i}$= standard population weight for age group $i$

Age adjustment allows meaningful comparison of cancer incidence between populations.

### 3.7.5 Kaplan–Meier Survival Estimation

The Kaplan–Meier method was used to estimate survival probability over time among cancer patients. This non-parametric method is commonly applied in medical research to analyze time-to-event data, particularly when some individuals are lost to follow-up or censored during the study period.

The Kaplan–Meier survival function is defined as:

$$\boldsymbol{S(t)} = \prod_{t_i \leq t} \left(\mathbf{1} - \frac{\boldsymbol{d_i}}{\boldsymbol{n_i}}\right)$$

Where:

$\boldsymbol{S(t)}$= probability of survival at time $t$

$\boldsymbol{d_i}$= number of deaths at time $t_i$

$\boldsymbol{n_i}$= number of individuals at risk just before time $t_i$

This method accounts for censored observations and provides a stepwise estimate of survival probability.

### 3.7.6 Log-Rank Test

The log-rank test was used to compare survival distributions between groups, particularly between male and female cancer patients. It is a widely used non-parametric statistical test for comparing survival curves derived from Kaplan–Meier analysis.

The test statistic for the log-rank test is calculated as:

$$\chi^2 = \frac{(O-E)^2}{V}$$

Where:

$O$= observed number of events

$E$= expected number of events

$V$= variance of the difference between observed and expected events

A significant p-value indicates that the survival curves differ between groups.

### 3.7.7 Cox Proportional Hazards Model

The Cox proportional hazards model was used to evaluate the effect of predictor variables such as age and sex on survival outcomes. This semi-parametric regression model is widely used in survival analysis to estimate the relationship between survival time and explanatory variables.

The hazard function for the Cox model is expressed as:

$$h(t \mid X) = h_0(t)e^{\beta X}$$

Where:

$h(t \mid X)$= hazard at time $t$given covariates $X$

$h_0(t)$= baseline hazard function

$\beta$= regression coefficient

$e^{\beta}$= hazard ratio

The hazard ratio represents the relative risk of death associated with a predictor variable.

### 3.8 Software Used for Analysis

All statistical analyses and data management procedures in this study were carried out using **Microsoft Excel, R statistical software, and SAS**. These software tools were used at

different stages of the analysis to ensure efficient data processing, accurate statistical computation, and clear graphical representation of results.

Initially, **Microsoft Excel** was used for preliminary data handling, including data entry, data cleaning, coding of variables, and organization of datasets obtained from Population-Based Cancer Registries. Excel was also used to generate basic descriptive summaries such as frequency distributions, percentages, and simple charts for exploratory analysis.

Further statistical analysis was performed using **R software**, which is widely used for epidemiological and statistical analysis. R was utilized to compute descriptive statistics, calculate epidemiological measures such as relative proportions, median age at diagnosis, age-specific incidence rates, and age-adjusted incidence rates. In addition, graphical visualizations such as trend plots and distribution charts were generated to illustrate patterns in cancer incidence and survival across different groups.

Advanced statistical procedures were conducted using **SAS software**. SAS procedures were applied for survival analysis, including Kaplan–Meier survival estimation, log-rank tests, and Cox proportional hazards regression modelling. SAS was also used for validation of analytical results and for handling large datasets efficiently.

### 3.9 Ethical Considerations

The study is based entirely on secondary data obtained from cancer registries. The data used for analysis were anonymized and aggregated to ensure patient confidentiality. No direct patient interaction was involved in the study.

### 3.10 Limitations of the Methodology

Although cancer registry data provide valuable epidemiological insights, certain limitations exist. Differences in reporting completeness, variations in diagnostic practices, and limited availability of certain clinical variables may influence the results. Additionally, survival estimates may be affected by differences in follow-up duration and healthcare accessibility across populations.

Despite these limitations, PBCR data remain an essential resource for studying cancer epidemiology and informing public health strategies.

# STATISTICAL ANALYSIS AND INTERPRETATION

## 4.1 Relative Proportion – Leading Cancer Sites (2015-19)

Relative proportion in epidemiology refers to the comparison of proportions of disease or health outcomes between different groups in order to determine the strength of association between exposure and disease.

### 4.1.1 Chennai – Seven Leading Sites of Cancer

***Table 4.1***

| Male | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Oral (15.8) | C00-C06 |
| Lung etc. (9.5) | C33-C34 |
| Stomach (7.6) | C16 |
| Prostate (7.2) | C61 |
| Colorectal (7.2) | C18-C20 |
| Oesophagus (3.5) | C15 |
| Brain, Nervous System (2.3) | C70-C72 |

***Table 4.2***

| Female | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Breast (33.3) | C50 |
| Cervix Uteri (8.9) | C53 |
| Ovary (5.8) | C56 |
| Colorectal (5.4) | C18-C20 |
| Oesophagus (2.3) | C15 |
| Thyroid (2.2) | C73 |
| Gall Bladder (1.3) | C23-C24 |

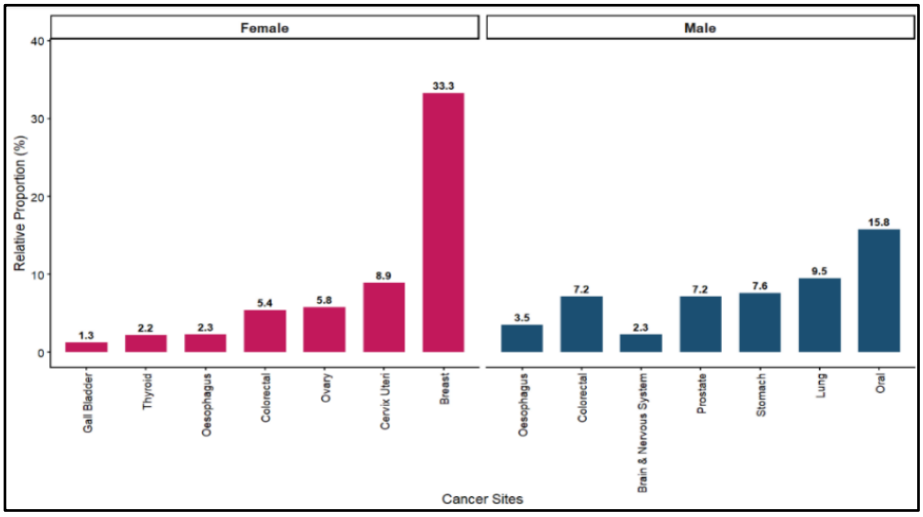



***Figure 4.1: Relative Proportion of Seven Leading Cancer Sites by Sex – Chennai (2015–2019)***

**Interpretation:**

During 2015–2019 in Chennai, clear sex-specific patterns are observed in leading cancer sites. Among males, oral cavity cancer (15.8%) is the most common, followed by lung (9.5%), stomach (7.6%), prostate (7.2%), and colorectal (7.2%) cancers, indicating a substantial burden of tobacco-related and gastrointestinal malignancies. Among females, breast cancer (33.3%) predominates, followed by cervix uteri (8.9%), ovary (5.8%), and colorectal (5.4%), reflecting the major contribution of reproductive and gastrointestinal cancers.

### 4.1.2 Bangalore – Seven Leading Sites of Cancer

*Table 4.3*

| Male | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Oral (10.1) | C00-C06 |
| Lung etc. (9.8) | C33-C34 |
| Colorectal (7.5) | C18-C20 |
| Prostate (7.0) | C61 |
| Stomach (6.6) | C16 |
| Oesophagus (4.4) | C15 |
| Brain, Nervous System (3.4) | C70-C72 |

*Table 4.4*

| Female | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Breast (31.6) | C50 |
| Cervix Uteri (9.1) | C53 |
| Ovary (6.4) | C56 |
| Colorectal (4.4) | C18-C20 |
| Thyroid (3.4) | C73 |
| Oesophagus (2.9) | C15 |
| Gall Bladder (1.2) | C23-C24 |

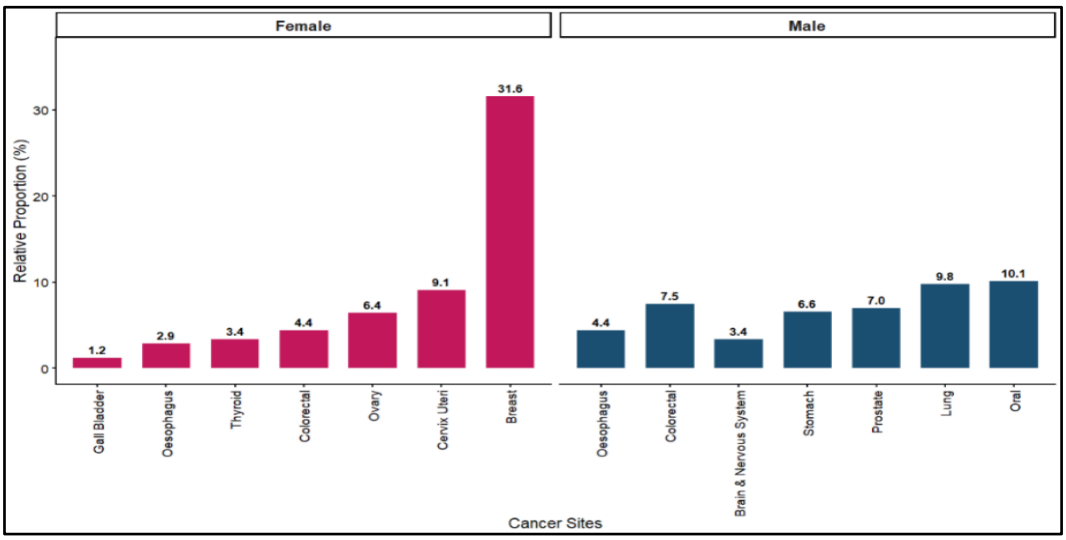



*Figure 4.2: Relative Proportion of Seven Leading Cancer Sites by Sex – Bangalore (2015–2019*

**Interpretation:**

During 2015–2019 in Bangalore, clear sex-specific patterns are observed in cancer distribution. Among males, oral cavity cancer (10.1%) is the leading site, followed by lung (9.8%), colorectal (7.5%), prostate (7.0%), and stomach (6.6%), reflecting the influence of tobacco use, ageing, and lifestyle factors. Among females, breast cancer (31.6%) predominates, followed by cervix uteri (9.1%), ovary (6.4%), and colorectal (4.4%), indicating the major contribution of reproductive and gastrointestinal cancers.

### 4.1.3 Delhi – Seven Leading Sites of Cancer (2015–2019)

***Table 4.5***

| Male | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Oral (14.6) | C00-C06 |
| Lung etc. (10.7) | C33-C34 |
| Prostate (7.1) | C61 |
| Colorectal (4.9) | C18-C20 |
| Oesophagus (3.7) | C15 |
| Brain, Nervous System (3.1) | C70-C72 |
| Stomach (2.7) | C16 |

***Table 4.6***

| Female | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Breast (27.1) | C50 |
| Cervix Uteri (8.7) | C53 |
| Gall Bladder (7.8) | C23-C24 |
| Ovary (6.42) | C56 |
| Colorectal (3.7) | C18-C20 |
| Oesophagus (2.2) | C15 |
| Thyroid (1.96) | C73 |

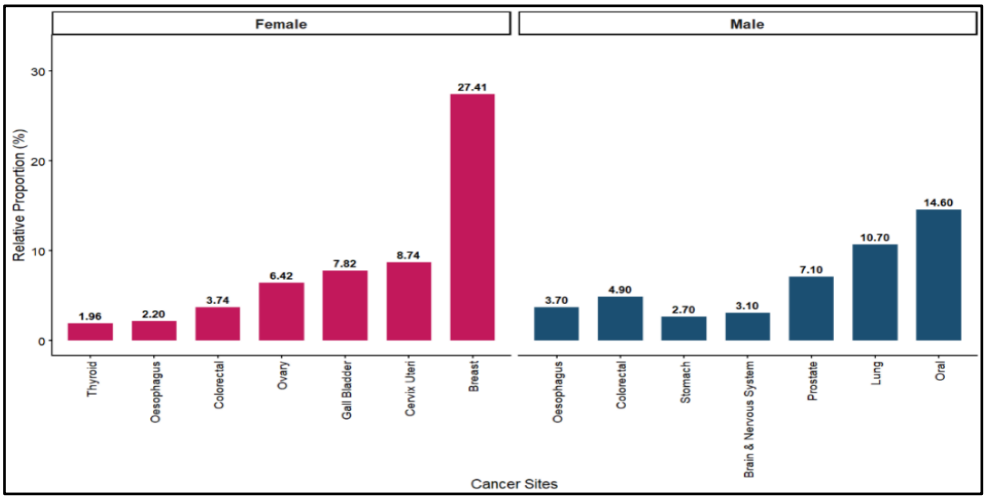



***Figure 4.3: Relative Proportion of Seven Leading Cancer Sites by Sex – Delhi (2015–2019)***

**Interpretation:**

During 2015–2019 in Delhi, clear sex-specific patterns are observed in cancer distribution. Among males, oral cavity (14.6%) and lung (10.7%) cancers are the leading sites, followed by prostate (7.1%), colorectal (4.9%), and oesophagus (3.7%), reflecting the influence of tobacco use, ageing, and lifestyle factors. Among females, breast cancer (27.1%) predominates, followed by cervix uteri (8.7%), gall bladder (7.8%), and ovary (6.4%), indicating the major contribution of breast cancer, persistent cervical cancer burden, and notable gall bladder malignancy.

### 4.1.4 Trivandrum – Seven Leading Sites of Cancer (2015–2019)

***Table 4.7***

| Male | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Lung etc. (12.9) | C33-C34 |
| Oral (10.03) | C00-C06 |
| Colorectal (8.09) | C18-C20 |
| Prostate (7.8) | C61 |
| Stomach (3.8) | C16 |
| Oesophagus (2.8) | C15 |
| Brain, Nervous System (1.9) | C70-C72 |

***Table 4.8***

| Female | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Breast (29.9) | C50 |
| Thyroid (11.0) | C73 |
| Colorectal (6.4) | C18-C20 |
| Ovary (5.2) | C56 |
| Cervix Uteri (4.8) | C53 |
| Gall Bladder (1.0) | C23-C24 |
| Oesophagus (0.8) | C15 |

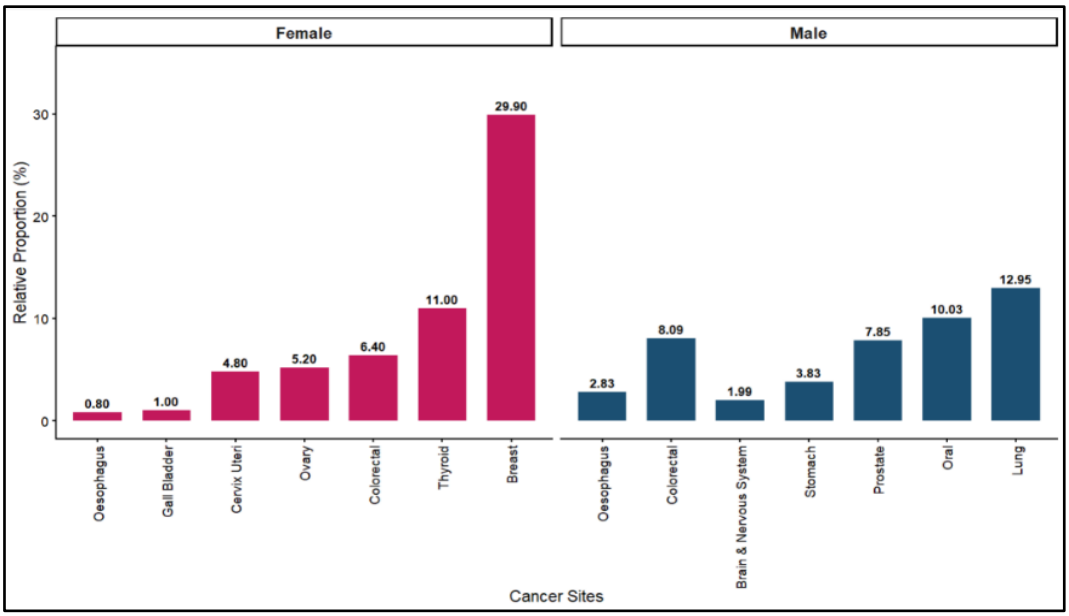



***Figure 4.4: Relative Proportion of Seven Leading Cancer Sites by Sex – Trivandrum (2015–2019)***

**Interpretation:**

During 2015–2019 in Trivandrum, clear sex-specific patterns are observed in the distribution of leading cancer sites. Among males, lung cancer (12.9%) is the leading site, followed by oral cavity (10.0%), colorectal (8.1%), and prostate (7.8%), reflecting the influence of tobacco use, ageing, and lifestyle factors. Among females, breast cancer (29.9%) predominates, followed by thyroid (11.0%), colorectal (6.4%), ovary (5.2%), and cervix uteri (4.8%), indicating the major contribution of breast and endocrine-related cancers.

### 4.1.5 Kollam – Seven Leading Sites of Cancer

***Table 4.9***

| Male | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Lung etc. (16.7) | C33-C34 |
| Oral (10.1) | C00-C06 |
| Colorectal (8.4) | C18-C20 |
| Prostate (6.28) | C61 |
| Stomach (4.5) | C16 |
| Oesophagus (3.4) | C15 |
| Brain, Nervous System (1.7) | C70-C72 |

***Table 4.10***

| Female | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Breast (30.6) | C50 |
| Thyroid (10.9) | C73 |
| Cervix Uteri (5.1) | C53 |
| Ovary (4.8) | C56 |
| Colorectal (4.5) | C18-C20 |
| Oesophagus (1.5) | C15 |
| Gall Bladder (1.0) | C23-C24 |

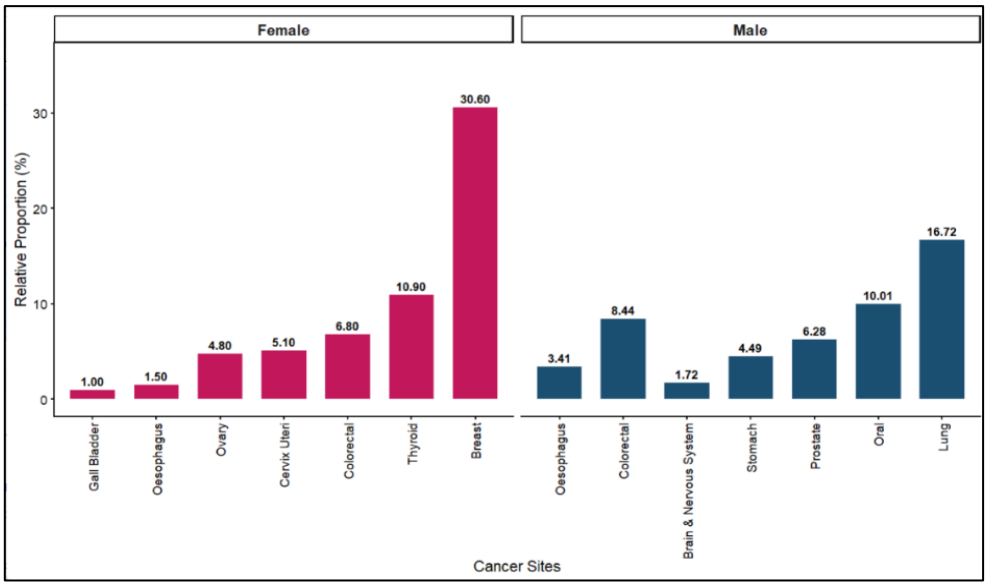



***Figure 4.5: Relative Proportion of Seven Leading Cancer Sites by Sex – Kollam (2015–2019)***

**Interpretation:**

During 2015–2019 in Kollam, clear sex-specific patterns are observed in the distribution of leading cancer sites. Among males, lung cancer (16.7%) is the leading site, followed by oral cavity (10.1%) and colorectal (8.4%), reflecting the impact of tobacco use, ageing, and lifestyle factors. Among females, breast cancer (30.6%) predominates, followed by thyroid (10.9%), cervix uteri (5.1%), ovary (4.8%), and colorectal (4.5%), highlighting the major contribution of breast and endocrine-related cancers.

### 4.1.6 Mumbai – Seven Leading Sites of Cancer (2015–2019)

***Table 4.10***

| Male | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Oral (15.6) | C00-C06 |
| Prostate (9.12) | C61 |
| Lung etc. (8.6) | C33-C34 |
| Colorectal (6.6) | C18-C20 |
| Brain, Nervous System (3.4) | C70-C72 |
| Stomach (3.8) | C16 |
| Oesophagus (3.2) | C15 |

***Table 4.11***

| Female | |
|---|---|
| **Sites – (%)** | **ICD-10** |
| Breast (21.8) | C50 |
| Cervix Uteri (17.2) | C53 |
| Oesophagus (6.2) | C15 |
| Ovary (5.8) | C56 |
| Colorectal (3.6) | C18-C20 |
| Thyroid (1.8) | C73 |
| Gall Bladder (1.5) | C23-C24 |

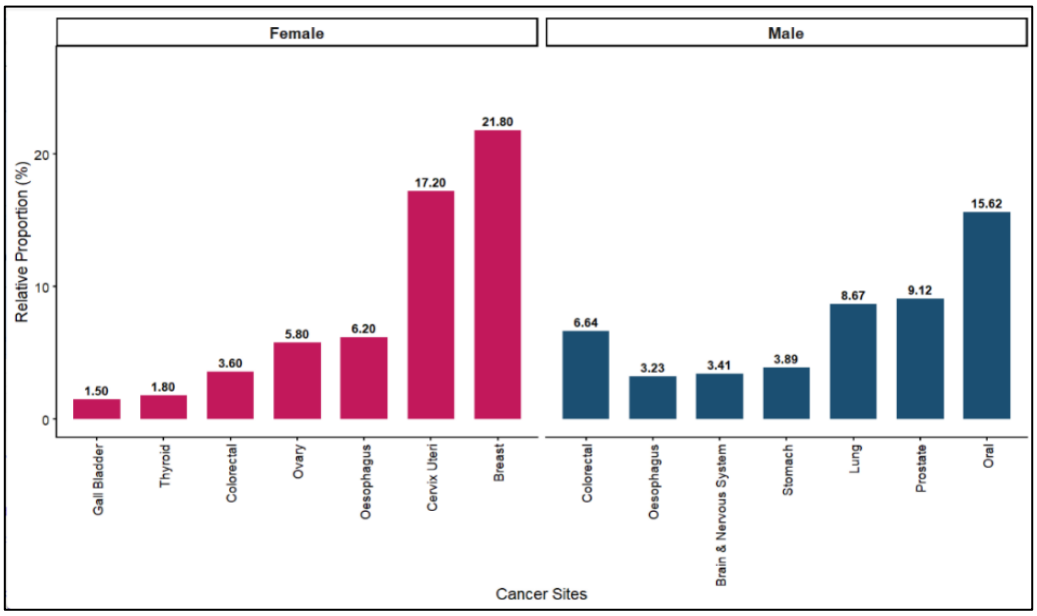



***Figure 4.6: Relative Proportion of Seven Leading Cancer Sites by Sex – Mumbai (2015–2019)***

**Interpretation:**

During 2015–2019 in Mumbai, clear sex-specific patterns are observed in the distribution of leading cancer sites. Among males, oral cavity cancer (15.6%) is the leading site, followed by prostate (9.1%), lung (8.6%), and colorectal (6.6%), indicating a substantial burden of tobacco-related cancers along with prostate cancer. Among females, breast cancer (21.8%) and cervix uteri (17.2%) are the leading sites, reflecting a dual burden of breast and cervical cancers, with ovary and oesophagus contributing moderate proportions.

## 4.2 Comparative Profile of Median Age and Incidence Indicators among Males across Population-Based Cancer Registries (PBCRs), 2015–2019.

***Table 4.12: Comparative Profile of Median Age and Incidence Indicators among Males across PBCRs (2015–2019)***

| Male | | | | |
|---|---|---|---|---|
| **Sl. No.** | **PBCRs** | **Adjusted Median Age** | **AAR (per 100,000)** | **ASPR** |
| 1 | Chennai | 62 | 99.6 | 226.5 |
| 2 | Delhi | 61 | 117.5 | 202.9 |
| 3 | Mumbai | 65 | 85.5 | 206.4 |
| 4 | Bangalore | 63 | 100.1 | 168.7 |
| 5 | Bhopal | 58 | 90.5 | 183.4 |
| 6 | Barshi Rural | 60 | 40.4 | 1502.2 |
| 7 | Thiruvananthapuram | 62 | 114.4 | 311.4 |
| 8 | Dibrugarh | 61 | 69.3 | 156.1 |
| 9 | Sikkim | 63 | 72.3 | 174.7 |
| 10 | Imphal West | 64 | 77.9 | 190.8 |
| 11 | Kamrup | 63 | 130.3 | 311.4 |
| 12 | Mizoram | 62 | 149.4 | 368.3 |
| 13 | Aurangabad | 58 | 449.0 | 920.3 |
| 14 | Nagpur | 56 | 72.0 | 143.1 |
| 15 | Kollam | 63 | 116.0 | 272.2 |
| 16 | Pune | 63 | 59.2 | 142.4 |

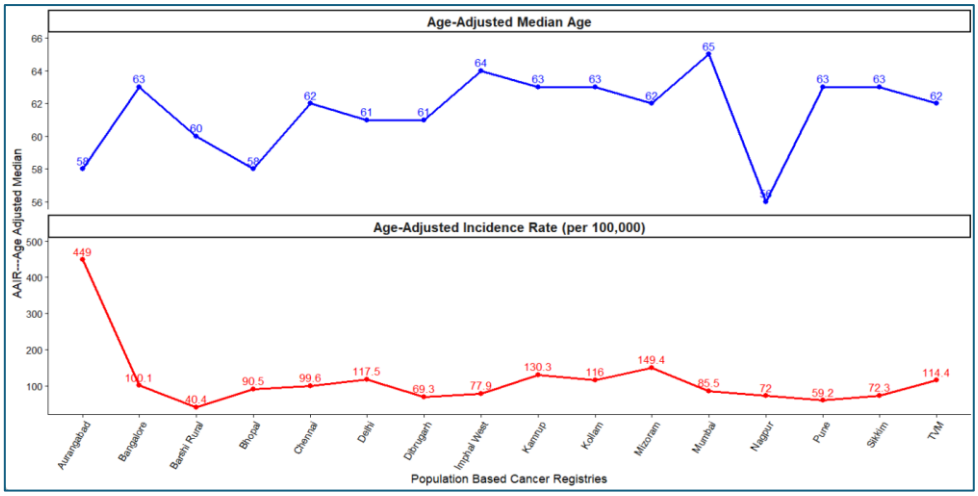



***Figure 4.7: Age-Adjusted Median Age and Age-Adjusted Incidence Rate among Males across PBCRs (2015–2019)***

**Interpretation – Males**

Across PBCRs, the age-adjusted median age at diagnosis among male ranges from 56 to 65 years, with most registries clustering between 60 and 64 years, indicating a relatively consistent age pattern of cancer occurrence. In contrast, the Age-Adjusted Incidence Rate (AAR) shows substantial regional variation, ranging from 40.4 per 100,000 in Barshi Rural to 449 per 100,000 in Aurangabad. As the rates are age-standardized, this variation cannot be explained by differences in population age structure. The stability of median age alongside wide disparities in incidence suggests that regional differences in cancer burden are likely influenced by variations in risk factor exposure, environmental conditions, healthcare access, and registry completeness.

## 4.3 Comparative Profile of Median Age and Incidence Indicators among Females across Population-Based Cancer Registries (PBCRs), 2015–2019

*Table 4.13: Comparative Profile of Median Age and Incidence Indicators among Females across PBCRs (2015–2019)*

| Female | | | | |
|---|---|---|---|---|
| **Sl. No.** | **PBCRs** | **Adjusted Median Age** | **AAR (per 100,000)** | **ASPR** |
| 1 | Chennai | 59 | 111.0 | 226.7 |
| 2 | Delhi | 56 | 107.1 | 210.8 |
| 3 | Mumbai | 60 | 90.7 | 180.1 |
| 4 | Bangalore | 58 | 121.3 | 250.7 |
| 5 | Bhopal | 55 | 88.9 | 166.5 |
| 6 | Barshi Rural | 54 | 50.4 | 81.7 |
| 7 | Thiruvananthapuram | 55 | 114.8 | 220.2 |
| 8 | Dibrugarh | 54 | 63.4 | 114.9 |
| 9 | Sikkim | 57 | 79.6 | 163.9 |
| 10 | Imphal West | 57 | 94.5 | 195.1 |
| 11 | Kamrup | 58 | 151.5 | 313.9 |
| 12 | Mizoram | 59 | 89.6 | 205.4 |
| 13 | Aurangabad | 55 | 368.1 | 695.3 |
| 14 | Nagpur | 52 | 72.5 | 126.0 |
| 15 | Kollam | 54 | 102.4 | 192.1 |
| 16 | Pune | 59 | 78.6 | 167.8 |

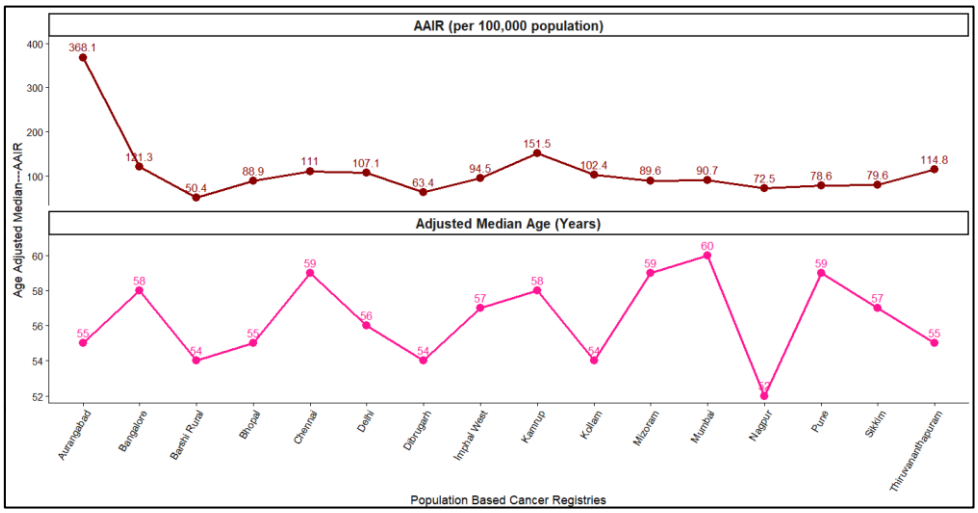



*Figure 4.8: AAIR and Adjusted Median Age among Females across PBCRs (2015–2019)*

**Interpretation – Females**

Across PBCRs, the age-adjusted incidence rate (AAIR) among males shows substantial regional variation, ranging from 50.4 per 100,000 in Barshi Rural to 368.1 per 100,000 in Aurangabad. In contrast, the adjusted median age at diagnosis varies within a narrow range of 52 to 60 years, with most registries clustering between 55 and 59 years. The relatively stable age distribution alongside marked differences in incidence suggests that regional disparities in cancer burden are unlikely to be explained by population age structure. Instead, these variations may reflect differences in risk factor prevalence, environmental exposures, healthcare accessibility, and the completeness of cancer registration systems.

## 4.4 Inter-Registry Variation in Age-Specific Incidence Rates by Sex across PBCRs, 2015–2019

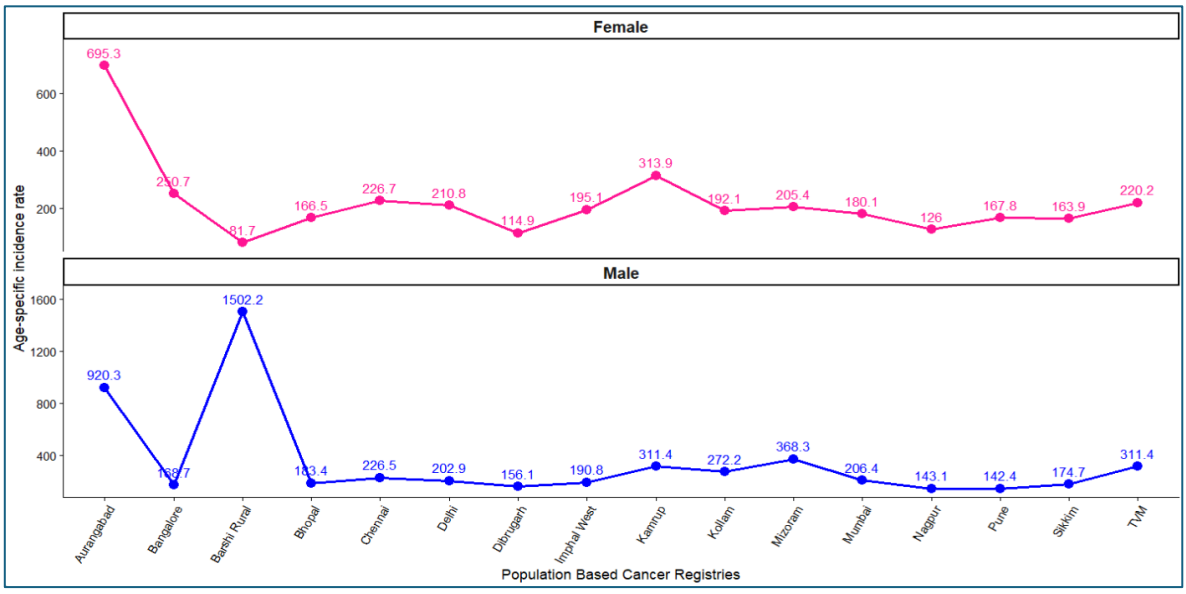



*Figure 4.9: Inter-Registry Variation in Age-Specific Incidence Rates by Sex across PBCRs (2015–2019)*

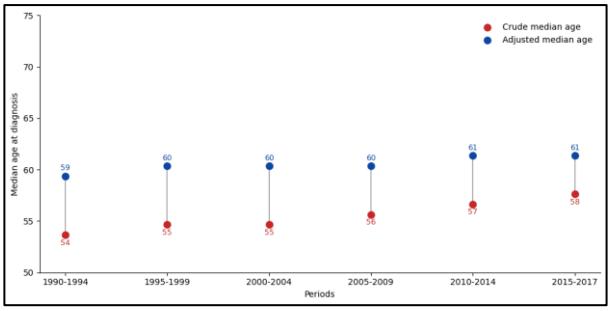
**Interpretation:**

Among females, cancer incidence rates vary widely across registries, from 81.7 in Barshi Rural to 695.3 in Aurangabad—an eight-fold difference. Elevated rates occur in Kamrup, Chennai, Thiruvananthapuram, Delhi, and Mizoram, while Dibrugarh is lower. Most registries fall between 160–230, with Aurangabad standing out as distinctly higher. Among males, variation is even greater, ranging from 142.4 in Pune to 1502.2 in Barshi Rural—a ten-fold difference. High rates are reported in Aurangabad, Mizoram, Kamrup, and

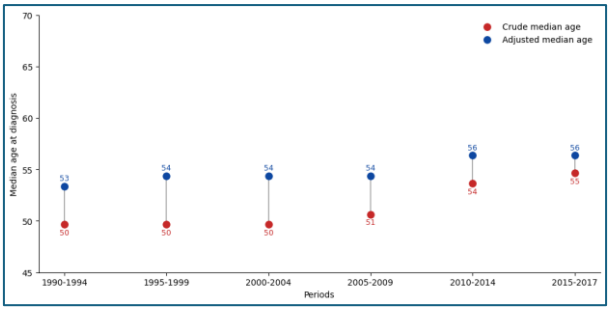
Thiruvananthapuram, while Bangalore, Dibrugarh, and Nagpur are lower. Overall, males show higher incidence than females, highlighting a marked sex differential in cancer burden.

### 4.5 Comparison of Crude and Age-Adjusted Median Age at Cancer Diagnosis over Six PBCRs

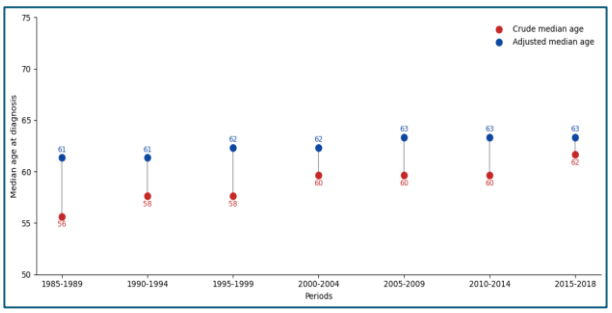
***Delhi PBCR – Males (1990–2017)***



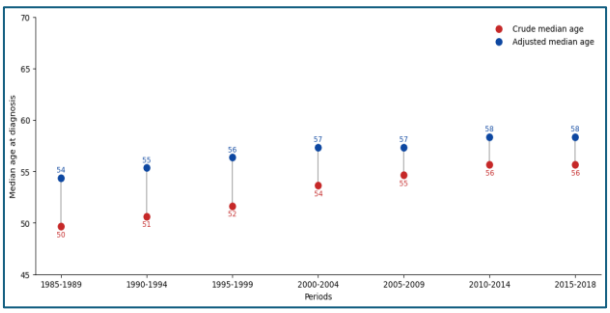

***Delhi PBCR – Females (1990–2017)***



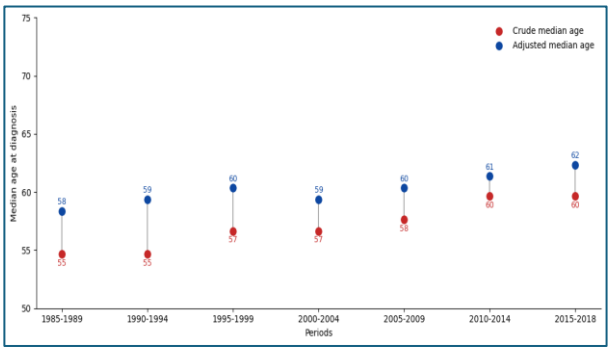

***Bangalore PBCR – Males (1985–2018)***



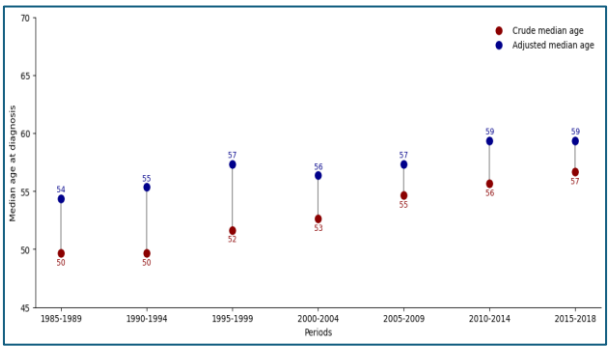

***Bangalore PBCR – Females (1985–2018)***



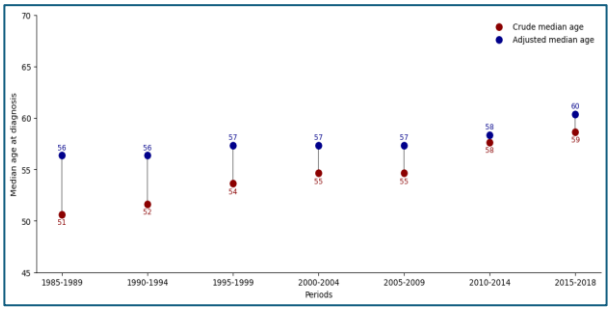

***Chennai PBCR – Males (1985–2018)***



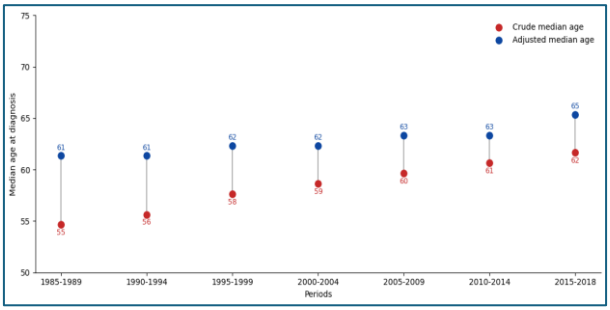

***Chennai PBCR – Females (1985–2018)***



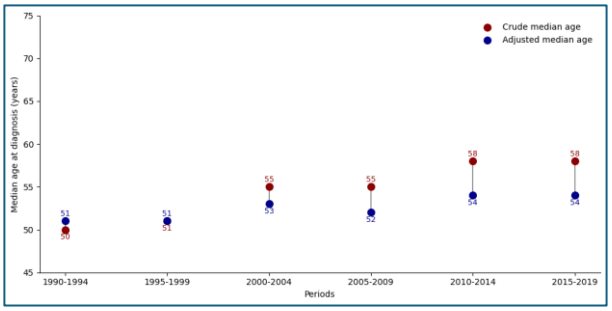
***Mumbai PBCR – Males (1985–2018)***



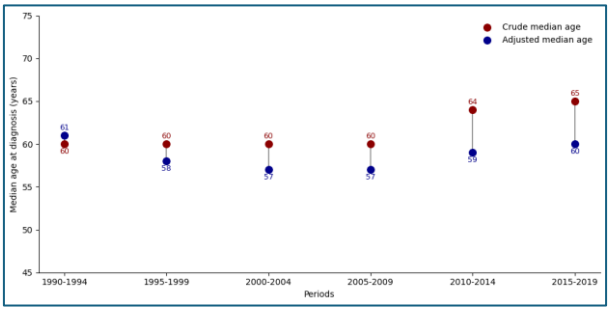

***Mumbai PBCR – Females (1985–2018)***



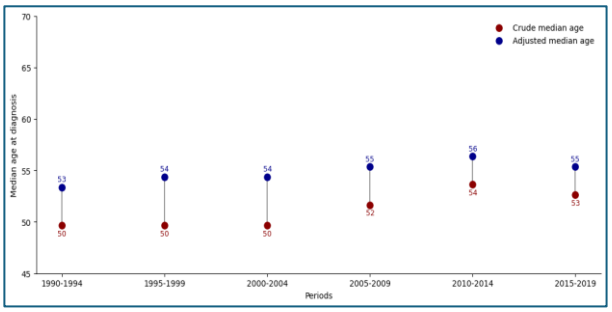

***Barshi PBCR – Males (1990–2019)***




***Barshi PBCR – Females (1990–2019)***




***Bhopal PBCR – Males (1990–2019)***




***Bhopal PBCR – Females (1990–2019)***

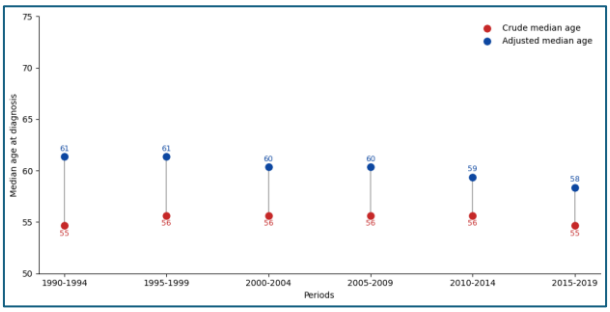



***Figure 4.10: Trend in Crude and Age-Adjusted Median Age of Cancer Cases across Selected PBCRs (1985–2019)***

*Table 4.14: Trend in Crude and Age-Adjusted Median Age of Cancer Cases across Selected PBCRs (1985–2019)*

| Registry | Period | Male Crude | Male Age-Adjusted | Female Crude | Female Age-Adjusted |
|---|---|---|---|---|---|
| Delhi | 1990–1994 | 54 | 59 | 50 | 53 |
| Delhi | 1995–1999 | 55 | 60 | 50 | 54 |
| Delhi | 2000–2004 | 55 | 60 | 50 | 54 |
| Delhi | 2005–2009 | 56 | 60 | 51 | 54 |
| Delhi | 2010–2014 | 57 | 61 | 54 | 56 |
| Delhi | 2015–2017 | 58 | 61 | 55 | 56 |
| Bangalore | 1985–1989 | 56 | 61 | 50 | 54 |
| Bangalore | 1990–1994 | 58 | 61 | 51 | 55 |
| Bangalore | 1995–1999 | 58 | 62 | 52 | 56 |
| Bangalore | 2000–2004 | 60 | 62 | 54 | 57 |
| Bangalore | 2005–2009 | 60 | 63 | 55 | 57 |
| Bangalore | 2010–2014 | 60 | 63 | 56 | 58 |
| Bangalore | 2015–2018 | 62 | 63 | 56 | 58 |
| Chennai | 1985–1989 | 55 | 58 | 50 | 54 |
| Chennai | 1990–1994 | 55 | 59 | 50 | 55 |
| Chennai | 1995–1999 | 57 | 60 | 52 | 57 |
| Chennai | 2000–2004 | 57 | 59 | 53 | 56 |
| Chennai | 2005–2009 | 58 | 60 | 55 | 57 |
| Chennai | 2010–2014 | 60 | 61 | 56 | 59 |
| Chennai | 2015–2018 | 60 | 62 | 57 | 59 |
| Mumbai | 1985–1989 | 55 | 61 | 51 | 56 |
| Mumbai | 1990–1994 | 56 | 61 | 52 | 56 |
| Mumbai | 1995–1999 | 58 | 62 | 54 | 57 |
| Mumbai | 2000–2004 | 59 | 62 | 55 | 57 |
| Mumbai | 2005–2009 | 60 | 63 | 55 | 57 |
| Mumbai | 2010–2014 | 61 | 63 | 58 | 58 |
| Mumbai | 2015–2018 | 62 | 65 | 59 | 60 |
| Barshi Rural | 1990–1994 | 60 | 61 | 50 | 51 |
| Barshi Rural | 1995–1999 | 60 | 58 | 51 | 51 |
| Barshi Rural | 2000–2004 | 60 | 57 | 55 | 53 |

| | | | | | |
|---|---|---|---|---|---|
| Barshi Rural | 2005–2009 | 60 | 57 | 55 | 52 |
| Barshi Rural | 2010–2014 | 64 | 59 | 58 | 54 |
| Barshi Rural | 2015–2019 | 65 | 60 | 58 | 54 |
| Bhopal | 1990–1994 | 55 | 61 | 50 | 53 |
| Bhopal | 1995–1999 | 56 | 61 | 50 | 54 |
| Bhopal | 2000–2004 | 56 | 60 | 50 | 54 |
| Bhopal | 2005–2009 | 56 | 60 | 52 | 55 |
| Bhopal | 2010–2014 | 56 | 59 | 54 | 56 |
| Bhopal | 2015–2019 | 55 | 58 | 53 | 55 |

**Interpretation:**

A longitudinal descriptive analysis of crude and age-adjusted median age at diagnosis across six Population-Based Cancer Registries (PBCRs) from 1985 to 2019 shows a consistent upward trend over time. Median age at diagnosis increased across most registries, with a stronger temporal rise observed in metropolitan regions (Delhi, Bangalore, Mumbai, Chennai) compared to Bhopal and Barshi Rural.

In Delhi, male crude median age increased from 54 to 58 years, with adjusted values consistently higher (59–61 years). Among females, crude age rose from 50 to 55 years, with adjusted values ranging from 53 to 56 years. Similar upward trends were observed in Bangalore, Chennai, and Mumbai, where both males and females showed steady increases in crude median age, while adjusted median age remained consistently higher and more stable.

In contrast, Bhopal showed relatively stable crude median age among males (55–56 years) with modest increases among females (50 to 53 years), while adjusted values ranged between 53 and 61 years. Barshi Rural exhibited a narrower gap between crude and adjusted estimates, with occasional overlap, indicating a different underlying age structure.

Overall, age-adjusted median age exceeded crude median age by approximately 2–6 years across most registries, highlighting the effect of age standardization. While crude median age reflects local demographic composition, age-adjusted median age provides a more comparable measure across populations.

### 4.6 Kaplan–Meier Survival Analysis

**Hypothesis for Kaplan–Meier**

**Null Hypothesis ($H_0$):**

There is no difference in the survival distributions between male and female patients.

**Alternative Hypothesis ($H_1$):**

There is a difference in survival distributions between male and female patients.

**Level of significance:** $\alpha = 0.05$

*Table 4.17: Test of Equality over Strata (Log-Rank Test)*

| Test of Equality over Strata | | | |
|---|---|---|---|
| **Test** | **Chi-Square** | **DF** | **Pr > Chi-Square** |
| Log-Rank | 5.8994 | 1 | 0.0151 |

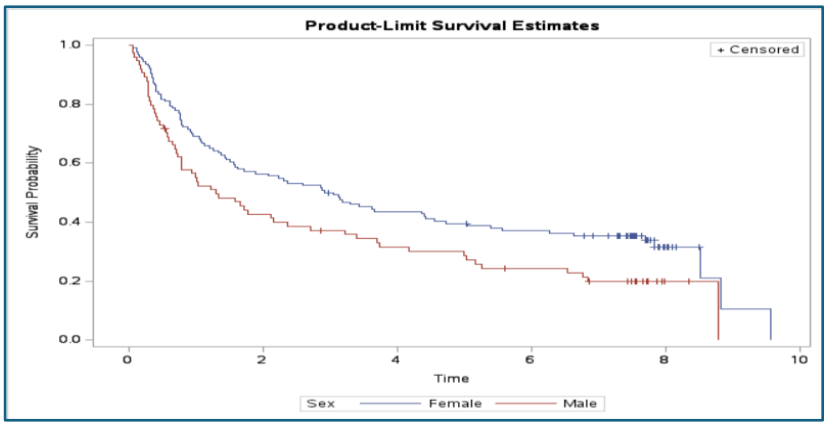



*Figure 4.11: Kaplan–Meier Product-Limit Survival Estimates by Sex*

**Interpretation:**

The Kaplan–Meier survival analysis demonstrated a gradual decline in survival probability over the follow-up period for both males and females. However, females consistently exhibited higher survival probabilities compared with males throughout the study period. The log-rank test showed a statistically significant difference between the survival curves ($\chi^2 = 5.899$, $p = 0.015$), indicating that the survival experiences of males and females are not equal. These results suggest that sex acts as an important prognostic factor in cancer survival, with female patients showing better survival outcomes than male patients during follow-up.

## 4.7 Cox Proportional Hazards Model

### Hypotheses for Cox Proportional Hazards

**Null Hypothesis ($H_0$):**

There is no association between the predictor variables and survival, and the hazard of death is the same regardless of sex and age.

**Alternative Hypothesis ($H_1$):**

At least one predictor variable (sex or age) is associated with survival, indicating that the hazard of death differs according to these variables.

**Level of significance:** $\alpha = 0.05$

***Table 4.18: Analysis of Maximum Likelihood Estimates***

| Analysis of Maximum Likelihood Estimates | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Parameter** | **DF** | **Parameter Estimate** | **Standard Error** | **Chi-Square** | **Pr > ChiSq** | **Hazard Ratio** | **Label** |
| Sex | 1 | -0.38696 | 0.17054 | 5.1488 | 0.0233 | 0.679 | Sex |
| Age | 1 | 0.01616 | 0.00643 | 6.3115 | 0.0120 | 1.016 | Age |

**Interpretation:**

The Cox proportional hazards model identified sex and age as significant predictors of survival. Females had a 32% lower risk of death than males ($\beta = -0.38696$, HR = 0.679, $p = 0.0233$). Age was also significant ($\beta = 0.01616$, HR = 1.016, $p = 0.0120$), indicating that each additional year increases the hazard of death by about 1.6%. These findings suggest that female patients have better survival outcomes, while increasing age is associated with poorer survival, likely due to higher comorbidity and reduced treatment tolerance among older patients.

# SUMMARY AND CONCLUSION

This study examined the epidemiological profile of cancer across selected Population-Based Cancer Registries (PBCRs) in India using statistical analysis of incidence patterns, median age at diagnosis, and survival outcomes. By integrating descriptive epidemiology with survival modeling techniques, the analysis provides insights into the distribution, determinants, and regional variability of cancer burden across populations.

The descriptive analysis of leading cancer sites demonstrated clear and consistent sex-specific patterns across multiple registries. Among males, cancers of the oral cavity, lung, colorectal region, and prostate were among the most frequently observed malignancies across several PBCRs. The predominance of oral and lung cancers across registries reflects the continued impact of tobacco consumption and associated behavioural risk factors in the Indian population. In contrast, among females, breast cancer was consistently identified as the leading cancer site across all registries included in the analysis. Other important malignancies among females included cancers of the cervix uteri, ovary, thyroid, and colorectal region. These patterns highlight the combined influence of reproductive, hormonal, lifestyle, and environmental factors in shaping female cancer profiles across regions.

A key statistical observation emerging from the study is the substantial inter-registry variation in cancer incidence indicators. The Age-Adjusted Incidence Rate (AAR) varied considerably across PBCRs, with some registries reporting relatively low incidence levels while others demonstrated substantially higher values. Such differences persisted even after age standardization, indicating that the variation cannot be attributed solely to differences in demographic age structures across populations. Instead, these variations likely reflect differences in exposure to risk factors, lifestyle patterns, environmental influences, healthcare accessibility, diagnostic infrastructure, and registry completeness.

Despite the marked variation in incidence rates, the analysis showed relatively limited variation in the adjusted median age at diagnosis across registries. Among males, the adjusted median age at cancer diagnosis generally ranged between approximately 56 and 65 years, with most registries clustered around 60–64 years. Among females, the adjusted median age ranged between approximately 52 and 60 years, with the majority of registries clustered between 54 and 59 years. This relatively narrow range indicates that the age distribution of cancer occurrence is broadly consistent across geographically diverse populations.

Longitudinal trend analysis further demonstrated a gradual increase in the median age at cancer diagnosis over successive time periods from 1985 to 2019. Metropolitan registries such as Delhi, Bangalore, Chennai, and Mumbai showed a more pronounced increase in median age compared with smaller or rural registries such as Barshi and Bhopal. This upward shift suggests that cancer occurrence is increasingly concentrated in older age groups, reflecting the effects of population ageing, improvements in life expectancy, and epidemiological transition in India.

Another important statistical finding relates to the comparison between crude and age-adjusted median age at diagnosis. Across most registries, the age-adjusted median age consistently exceeded the crude median age by approximately two to six years. This difference highlights the statistical importance of age standardization when comparing cancer indicators across populations with varying demographic structures. Crude measures reflect the underlying age composition of the population, whereas age-adjusted measures provide a standardized estimate that allows meaningful comparisons across registries.

The survival analysis component of the study provided additional insights into prognostic determinants of cancer outcomes. Kaplan–Meier survival estimation demonstrated that survival probability declined progressively over the follow-up period among both males and females. However, females consistently exhibited higher survival probabilities compared with males. The log-rank test indicated that the difference in survival distributions between sexes was statistically significant, suggesting that sex plays an important role in influencing cancer prognosis.

Further evaluation using the Cox proportional hazards regression model confirmed that both sex and age were statistically significant predictors of survival outcomes. The estimated hazard ratio for sex indicated that female patients had a substantially lower risk of mortality compared with male patients. Specifically, females exhibited approximately a 32 percent lower hazard of death relative to males. Age was also found to be a significant predictor, with each additional year of age associated with a modest but measurable increase in the hazard of mortality. These results underscore the importance of demographic and biological factors in determining survival outcomes among cancer patients.

From a statistical perspective, the findings highlight the value of integrating multiple epidemiological indicators—including relative proportions, age-adjusted incidence rates, median age measures, and survival analysis—to obtain a comprehensive understanding of

cancer patterns across populations. Descriptive statistics provided insights into the distribution of cancer sites, while inferential survival models helped identify prognostic factors influencing patient outcomes.

From a public health perspective, the study provides several important insights relevant to cancer control and prevention strategies in India. The predominance of oral and lung cancers among males strongly underscores the continued burden of tobacco-related malignancies. Tobacco consumption remains one of the most significant modifiable risk factors for cancer in India, and the findings reinforce the need for sustained public health interventions aimed at tobacco control. Strengthening tobacco cessation programs, enforcing regulatory policies on tobacco use, and expanding awareness campaigns remain critical priorities for reducing the burden of these preventable cancers.

Among females, the dominance of breast cancer across all registries highlights the increasing importance of breast cancer as a major public health concern. The findings suggest a growing need for strengthened screening, early detection, and awareness programs focused on breast health. In addition, the continued presence of cervical cancer among leading cancer sites indicates that cervical cancer prevention strategies—including HPV vaccination and cervical screening programs—remain essential components of national cancer control efforts.

The wide variation in cancer incidence rates across registries also suggests disparities in risk factor exposure, healthcare infrastructure, and diagnostic capacity across regions. Some registries report substantially higher incidence rates, which may reflect a combination of increased risk factor prevalence, better diagnostic detection, and improved cancer registration systems. Conversely, lower incidence rates in certain regions may partly reflect under-diagnosis or incomplete reporting. Strengthening cancer surveillance systems and improving the quality and coverage of PBCRs will therefore be essential for accurately monitoring cancer trends and guiding public health policies.

The upward trend in median age at cancer diagnosis observed in this study also has important implications for healthcare planning. As cancer incidence becomes increasingly concentrated in older age groups, healthcare systems will need to prepare for a growing demand for cancer care services among ageing populations. This includes strengthening geriatric oncology services, improving access to diagnostic and treatment facilities, and integrating cancer care into broader non-communicable disease control strategies.

Overall, the study demonstrates that cancer epidemiology in India is characterized by substantial regional variation, strong sex-specific patterns in cancer types, and significant demographic influences on survival outcomes. While the age distribution of cancer diagnosis remains relatively stable across populations, incidence rates vary widely across regions, reflecting the complex interplay of behavioural, environmental, and healthcare-related factors.

In conclusion, the findings emphasize the importance of evidence-based cancer control strategies that integrate epidemiological surveillance, risk factor reduction, early detection programs, and improved access to treatment. Continued strengthening of population-based cancer registries will be essential for monitoring trends, identifying high-risk populations, and informing effective public health interventions. A coordinated approach involving prevention, screening, early diagnosis, and treatment will be critical for reducing the growing burden of cancer and improving survival outcomes in the Indian population.

## COMPARISON WITH GLOBAL EVIDENCE

Similar patterns have been consistently observed in international studies, where the median age at cancer diagnosis undergoes a noticeable shift after age standardisation. This adjustment accounts for differences in underlying population age structures, allowing for more valid comparisons across countries and regions. The findings of the present study are therefore in alignment with global evidence and reflect broader demographic transitions as well as established principles in cancer epidemiology.

Age standardisation plays a critical role in interpreting median age at diagnosis, as crude estimates are heavily influenced by the age distribution of the population. In this study, the median age at diagnosis changed by as much as 10 years after standardisation, indicating that population structure is a major confounding factor. Without adjustment, populations with younger age distributions tend to show artificially lower median ages, while populations with older age structures exhibit higher median ages, even if the biological onset of disease is similar.

A clear divergence was observed between low- and middle-income countries (LMICs) and high-income countries (HICs). In LMICs, age standardisation generally resulted in an increase in the median age at diagnosis. This is primarily because these populations are demographically younger, and crude estimates are skewed toward younger age groups. Once

adjusted to a standard population, the underlying age pattern of cancer becomes more apparent, shifting the median upward. In contrast, in HICs, where populations are older, age standardisation often led to a decrease in median age, reflecting the removal of the effect of population ageing.

Furthermore, site-specific differences in age at diagnosis highlight important epidemiological variations. Colon cancer tends to be diagnosed at younger ages in LMICs, which may be attributed to differences in dietary patterns, environmental exposures, genetic susceptibility, and limited screening coverage. On the other hand, prostate cancer is typically diagnosed at older ages in LMICs compared with HICs, likely due to lower levels of early detection and screening practices such as prostate-specific antigen (PSA) testing.

These contrasting patterns underscore substantial heterogeneity in cancer epidemiology across regions. Variations in risk factor exposure, including lifestyle behaviours (diet, tobacco use, physical activity), environmental influences, and occupational hazards, contribute to differences in disease onset. In addition, disparities in healthcare infrastructure, access to diagnostic services, and implementation of screening programs further influence the age at which cancers are detected.

Overall, the findings emphasize that age standardisation is not merely a statistical adjustment but a necessary methodological step to ensure meaningful comparisons across populations. It allows researchers to disentangle the effects of demographic structure from true epidemiological differences, thereby providing a clearer understanding of global cancer patterns and supporting evidence-based public health planning (Zahed et al., 2024).

**Limitation:**

The study has several limitations. Possible misclassification of age and under-reporting in early registry years may have biased the estimates. The absence of individual-level data on risk factors and screening limits causal interpretation. Additionally, the choice of standard population may influence adjusted results. Finally, findings may not be fully generalizable to India, and inclusion of more PBCRs is needed to better capture regional variations in median age at diagnosis.

**Strengths**

A major strength of this study is the use of long-term, high-quality population-based cancer registry data spanning 1985–2019, which enables robust assessment of temporal trends. The application of age-adjusted median age provides a more accurate and comparable measure by accounting for differences in population age structures over time and across regions. Additionally, the inclusion of multiple Population-Based Cancer Registries allows for meaningful inter-regional comparisons within India, enhancing the reliability and generalizability of the findings.

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